

## SECTION 7

### 8-BIT INSTRUCTION EXECUTION TIMES

This section contains listings of the instruction execution times in terms of external clock (CLK) periods for the MC68008 and MC68HC001/MC68EC000 in 8-bit mode. In this data, it is assumed that both memory read and write cycles consist of four clock periods. A longer memory cycle causes the generation of wait states that must be added to the total instruction times.

The number of bus read and write cycles for each instruction is also included with the timing data. This data is shown as

$$n(r/w)$$

where:

n is the total number of clock periods

r is the number of read cycles

w is the number of write cycles

For example, a timing number shown as 18(3/1) means that 18 clock periods are required to execute the instruction. Of the 18 clock periods, 12 are used for the three read cycles (four periods per cycle). Four additional clock periods are used for the single write cycle, for a total of 16 clock periods. The bus is idle for two clock periods during which the processor completes the internal operations required for the instruction.

#### NOTE

The total number of clock periods (n) includes instruction fetch and all applicable operand fetches and stores.

#### 7.1 OPERAND EFFECTIVE ADDRESS CALCULATION TIMES

Table 7-1 lists the numbers of clock periods required to compute the effective addresses for instructions. The totals include fetching any extension words, computing the address, and fetching the memory operand. The total number of clock periods, the number of read cycles, and the number of write cycles (zero for all effective address calculations) are shown in the previously described format.

**Table 7-1. Effective Address Calculation Times**

| Addressing Mode |                                              | Byte    | Word    | Long    |
|-----------------|----------------------------------------------|---------|---------|---------|
| Register        |                                              |         |         |         |
| Dn              | Data Register Direct                         | 0(0/0)  | 0(0/0)  | 0(0/0)  |
| An              | Address Register Direct                      | 0(0/0)  | 0(0/0)  | 0(0/0)  |
| Memory          |                                              |         |         |         |
| (An)            | Address Register Indirect                    | 4(1/0)  | 8(2/0)  | 16(4/0) |
| (An)+           | Address Register Indirect with Postincrement | 4(1/0)  | 8(2/0)  | 16(4/0) |
| -(An)           | Address Register Indirect with Predecrement  | 6(1/0)  | 10(2/0) | 18(4/0) |
| (d16, An)       | Address Register Indirect with Displacement  | 12(3/0) | 16(4/0) | 24(6/0) |
| (dg, An, Xn)*   | Address Register Indirect with Index         | 14(3/0) | 18(4/0) | 26(6/0) |
| (xxx).W         | Absolute Short                               | 12(3/0) | 16(4/0) | 24(6/0) |
| (xxx).L         | Absolute Long                                | 20(5/0) | 24(6/0) | 32(8/0) |
| (d16, PC)       | Program Counter Indirect with Displacement   | 12(3/0) | 16(3/0) | 24(6/0) |
| (dg, PC, Xn)*   | Program Counter Indirect with Index          | 14(3/0) | 18(4/0) | 26(6/0) |
| #<data>         | Immediate                                    | 8(2/0)  | 8(2/0)  | 16(4/0) |

\*The size of the index register (Xn) does not affect execution time.

## 7.2 MOVE INSTRUCTION EXECUTION TIMES

Tables 7-2, 7-3, and 7-4 list the numbers of clock periods for the move instructions. The totals include instruction fetch, operand reads, and operand writes. The total number of clock periods, the number of read cycles, and the number of write cycles are shown in the previously described format.

**Table 7-2. Move Byte Instruction Execution Times**

| Source        | Destination |         |         |         |         |           |               |         |          |
|---------------|-------------|---------|---------|---------|---------|-----------|---------------|---------|----------|
|               | Dn          | An      | (An)    | (An)+   | -(An)   | (d16, An) | (dg, An, Xn)* | (xxx).W | (xxx).L  |
| Dn            | 8(2/0)      | 8(2/0)  | 12(2/1) | 12(2/1) | 12(2/1) | 20(4/1)   | 22(4/1)       | 20(4/1) | 28(6/1)  |
| An            | 8(2/0)      | 8(2/0)  | 12(2/1) | 12(2/1) | 12(2/1) | 20(4/1)   | 22(4/1)       | 20(4/1) | 28(6/1)  |
| (An)          | 12(3/0)     | 12(3/0) | 16(3/1) | 16(3/1) | 16(3/1) | 24(5/1)   | 26(5/1)       | 24(5/1) | 32(7/1)  |
| (An)+         | 12(3/0)     | 12(3/0) | 16(3/1) | 16(3/1) | 16(3/1) | 24(5/1)   | 26(5/1)       | 24(5/1) | 32(7/1)  |
| -(An)         | 14(3/0)     | 14(3/0) | 18(3/1) | 18(3/1) | 18(3/1) | 26(5/1)   | 28(5/1)       | 26(5/1) | 34(7/1)  |
| (d16, An)     | 20(5/0)     | 20(5/0) | 24(5/1) | 24(5/1) | 24(5/1) | 32(7/1)   | 34(7/1)       | 32(7/1) | 40(9/1)  |
| (dg, An, Xn)* | 22(5/0)     | 22(5/0) | 26(5/1) | 26(5/1) | 26(5/1) | 34(7/1)   | 36(7/1)       | 34(7/1) | 42(9/1)  |
| (xxx).W       | 20(5/0)     | 20(5/0) | 24(5/1) | 24(5/1) | 24(5/1) | 32(7/1)   | 34(7/1)       | 32(7/1) | 40(9/1)  |
| (xxx).L       | 28(7/0)     | 28(7/0) | 32(7/1) | 32(7/1) | 32(7/1) | 40(9/1)   | 42(9/1)       | 40(9/1) | 48(11/1) |
| (d16, PC)     | 20(5/0)     | 20(5/0) | 24(5/1) | 24(5/1) | 24(5/1) | 32(7/1)   | 34(7/1)       | 32(7/1) | 40(9/1)  |
| (dg, PC, Xn)* | 22(5/0)     | 22(5/0) | 26(5/1) | 26(5/1) | 26(5/1) | 34(7/1)   | 36(7/1)       | 34(7/1) | 42(9/1)  |
| #<data>       | 16(4/0)     | 16(4/0) | 20(4/1) | 20(4/1) | 20(4/1) | 28(6/1)   | 30(6/1)       | 28(6/1) | 36(8/1)  |

\*The size of the index register (Xn) does not affect execution time.

**Table 7-3. Move Word Instruction Execution Times**

| Source                     | Destination |         |         |         |         |                        |                            |          |          |
|----------------------------|-------------|---------|---------|---------|---------|------------------------|----------------------------|----------|----------|
|                            | Dn          | An      | (An)    | (An)+   | -(An)   | (d <sub>16</sub> , An) | (d <sub>8</sub> , An, Xn)* | (xxx).W  | (xxx).L  |
| Dn                         | 8(2/0)      | 8(2/0)  | 16(2/2) | 16(2/2) | 16(2/2) | 24(4/2)                | 26(4/2)                    | 24(4/2)  | 32(6/2)  |
| An                         | 8(2/0)      | 8(2/0)  | 16(2/2) | 16(2/2) | 16(2/2) | 24(4/2)                | 26(4/2)                    | 24(4/2)  | 32(6/2)  |
| (An)                       | 16(4/0)     | 16(4/0) | 24(4/2) | 24(4/2) | 24(4/2) | 32(6/2)                | 34(6/2)                    | 32(6/2)  | 40(8/2)  |
| (An)+                      | 16(4/0)     | 16(4/0) | 24(4/2) | 24(4/2) | 24(4/2) | 32(6/2)                | 34(6/2)                    | 32(6/2)  | 40(8/2)  |
| -(An)                      | 18(4/0)     | 18(4/0) | 26(4/2) | 26(4/2) | 26(4/2) | 34(6/2)                | 32(6/2)                    | 34(6/2)  | 42(8/2)  |
| (d <sub>16</sub> , An)     | 24(6/0)     | 24(6/0) | 32(6/2) | 32(6/2) | 32(6/2) | 40(8/2)                | 42(8/2)                    | 40(8/2)  | 48(10/2) |
| (d <sub>8</sub> , An, Xn)* | 26(6/0)     | 26(6/0) | 34(6/2) | 34(6/2) | 34(6/2) | 42(8/2)                | 44(8/2)                    | 42(8/2)  | 50(10/2) |
| (xxx).W                    | 24(6/0)     | 24(6/0) | 32(6/2) | 32(6/2) | 32(6/2) | 40(8/2)                | 42(8/2)                    | 40(8/2)  | 48(10/2) |
| (xxx).L                    | 32(8/0)     | 32(8/0) | 40(8/2) | 40(8/2) | 40(8/2) | 48(10/2)               | 50(10/2)                   | 48(10/2) | 56(12/2) |
| (d <sub>16</sub> , PC)     | 24(6/0)     | 24(6/0) | 32(6/2) | 32(6/2) | 32(6/2) | 40(8/2)                | 42(8/2)                    | 40(8/2)  | 48(10/2) |
| (d <sub>8</sub> , PC, Xn)* | 26(6/0)     | 26(6/0) | 34(6/2) | 34(6/2) | 34(6/2) | 42(8/2)                | 44(8/2)                    | 42(8/2)  | 50(10/2) |
| #<data>                    | 16(4/0)     | 16(4/0) | 24(4/2) | 24(4/2) | 24(4/2) | 32(6/2)                | 34(6/2)                    | 32(6/2)  | 40(8/2)  |

\*The size of the index register (Xn) does not affect execution time.

**Table 7-4. Move Long Instruction Execution Times**

| Source                     | Destination |          |          |          |          |                        |                            |          |          |
|----------------------------|-------------|----------|----------|----------|----------|------------------------|----------------------------|----------|----------|
|                            | Dn          | An       | (An)     | (An)+    | -(An)    | (d <sub>16</sub> , An) | (d <sub>8</sub> , An, Xn)* | (xxx).W  | (xxx).L  |
| Dn                         | 8(2/0)      | 8(2/0)   | 24(2/4)  | 24(2/4)  | 24(2/4)  | 32(4/4)                | 34(4/4)                    | 32(4/4)  | 40(6/4)  |
| An                         | 8(2/0)      | 8(2/0)   | 24(2/4)  | 24(2/4)  | 24(2/4)  | 32(4/4)                | 34(4/4)                    | 32(4/4)  | 40(6/4)  |
| (An)                       | 24(6/0)     | 24(6/0)  | 40(6/4)  | 40(6/4)  | 40(6/4)  | 48(8/4)                | 50(8/4)                    | 48(8/4)  | 56(10/4) |
| (An)+                      | 24(6/0)     | 24(6/0)  | 40(6/4)  | 40(6/4)  | 40(6/4)  | 48(8/4)                | 50(8/4)                    | 48(8/4)  | 56(10/4) |
| -(An)                      | 26(6/0)     | 26(6/0)  | 42(6/4)  | 42(6/4)  | 42(6/4)  | 50(8/4)                | 52(8/4)                    | 50(8/4)  | 58(10/4) |
| (d <sub>16</sub> , An)     | 32(8/0)     | 32(8/0)  | 48(8/4)  | 48(8/4)  | 48(8/4)  | 56(10/4)               | 58(10/4)                   | 56(10/4) | 64(12/4) |
| (d <sub>8</sub> , An, Xn)* | 34(8/0)     | 34(8/0)  | 50(8/4)  | 50(8/4)  | 50(8/4)  | 58(10/4)               | 60(10/4)                   | 58(10/4) | 66(12/4) |
| (xxx).W                    | 32(8/0)     | 32(8/0)  | 48(8/4)  | 48(8/4)  | 48(8/4)  | 56(10/4)               | 58(10/4)                   | 56(10/4) | 64(12/4) |
| (xxx).L                    | 40(10/0)    | 40(10/0) | 56(10/4) | 56(10/4) | 56(10/4) | 64(12/4)               | 66(12/4)                   | 64(12/4) | 72(14/4) |
| (d <sub>16</sub> , PC)     | 32(8/0)     | 32(8/0)  | 48(8/4)  | 48(8/4)  | 48(8/4)  | 56(10/4)               | 58(10/4)                   | 56(10/4) | 64(12/4) |
| (d <sub>8</sub> , PC, Xn)* | 34(8/0)     | 34(8/0)  | 50(8/4)  | 50(8/4)  | 50(8/4)  | 58(10/4)               | 60(10/4)                   | 58(10/4) | 66(12/4) |
| #<data>                    | 24(6/0)     | 24(6/0)  | 40(6/4)  | 40(6/4)  | 40(6/4)  | 48(8/4)                | 50(8/4)                    | 48(8/4)  | 56(10/4) |

\*The size of the index register (Xn) does not affect execution time.

## 7.3 STANDARD INSTRUCTION EXECUTION TIMES

The numbers of clock periods shown in Table 7-5 indicate the times required to perform the operations, store the results, and read the next instruction. The total number of clock periods, the number of read cycles, and the number of write cycles are shown in the previously described format. The number of clock periods, the number of read cycles, and the number of write cycles, respectively, must be added to those of the effective address calculation where indicated by a plus sign (+).

In Table 7-5, the following notation applies:

- An — Address register operand
- Dn — Data register operand
- ea — An operand specified by an effective address
- M — Memory effective address operand

**Table 7-5. Standard Instruction Execution Times**

| Instruction | Size  | op<ea>, An | op<ea>, Dn  | op Dn, <M> |
|-------------|-------|------------|-------------|------------|
| ADD/ADDA    | Byte  | —          | 8(2/0)+     | 12(2/1)+   |
|             | Word  | 12(2/0)+   | 8(2/0)+     | 16(2/2)+   |
|             | Long  | 10(2/0)+** | 10(2/0)+**  | 24(2/4)+   |
| AND         | Byte  | —          | 8(2/0)+     | 12(2/1)+   |
|             | Word  | —          | 8(2/0)+     | 16(2/2)+   |
|             | Long  | —          | 10(2/0)+**  | 24(2/4)+   |
| CMP/CMPL    | Byte  | —          | 8(2/0)+     | —          |
|             | Word  | 10(2/0)+   | 8(2/0)+     | —          |
|             | Long  | 10(2/0)+   | 10(2/0)+    | —          |
| DIVS        | —     | —          | 162(2/0)+*  | —          |
| DIVU        | —     | —          | 144(2/0)+*  | —          |
| EOR         | Byte, | —          | 8(2/0)+***  | 12(2/1)+   |
|             | Word, | —          | 8(2/0)+***  | 16(2/2)+   |
|             | Long  | —          | 12(2/0)+*** | 24(2/4)+   |
| MULS        | —     | —          | 74(2/0)+*   | —          |
| MULU        | —     | —          | 74(2/0)+*   | —          |
| OR          | Byte, | —          | 8(2/0)+     | 12(2/1)+   |
|             | Word  | —          | 8(2/0)+     | 16(2/2)+   |
|             | Long  | —          | 10(2/0)+**  | 24(2/4)+   |
| SUB         | Byte, | —          | 8(2/0)+     | 12(2/1)+   |
|             | Word  | 12(2/0)+   | 8(2/0)+     | 16(2/2)+   |
|             | Long  | 10(2/0)+** | 10(2/0)+**  | 24(2/4)+   |

+ Add effective address calculation time.

\* Indicates maximum base value added to word effective address time

\*\* The base time of 10 clock periods is increased to 12 if the effective address mode is register direct or immediate (effective address time should also be added).

\*\*\* Only available effective address mode is data register direct.

DIVS, DIVU — The divide algorithm used by the MC68008 provides less than 10% difference between the best- and worst-case timings.

MULS, MULU — The multiply algorithm requires  $42+2n$  clocks where  $n$  is defined as:  
 MULS:  $n$  = tag the <ea> with a zero as the MSB;  $n$  is the resultant number of 10 or 01 patterns in the 17-bit source; i.e., worst case happens when the source is \$5555.  
 MULU:  $n$  = the number of ones in the <ea>

## 7.4 IMMEDIATE INSTRUCTION EXECUTION TIMES

The numbers of clock periods shown in Table 7-6 include the times to fetch immediate operands, perform the operations, store the results, and read the next operation. The total number of clock periods, the number of read cycles, and the number of write cycles are

shown in the previously described format. The number of clock periods, the number of read cycles, and the number of write cycles, respectively, must be added to those of the effective address calculation where indicated by a plus sign (+).

In Table 7-6, the following notation applies:

- # — Immediate operand
- Dn — Data register operand
- An — Address register operand
- M — Memory operand

**Table 7-6. Immediate Instruction Execution Times**

| Instruction | Size | op #, Dn | op #, An | op #, M  |
|-------------|------|----------|----------|----------|
| ADDI        | Byte | 16(4/0)  | —        | 20(4/1)+ |
|             | Word | 16(4/0)  | —        | 24(4/2)+ |
|             | Long | 28(6/0)  | —        | 40(6/4)+ |
| ADDQ        | Byte | 8(2/0)   | —        | 12(2/1)+ |
|             | Word | 8(2/0)   | 12(2/0)  | 16(2/2)+ |
|             | Long | 12(2/0)  | 12(2/0)  | 24(2/4)+ |
| ANDI        | Byte | 16(4/0)  | —        | 20(4/1)+ |
|             | Word | 16(4/0)  | —        | 24(4/2)+ |
|             | Long | 28(6/0)  | —        | 40(6/4)+ |
| CMPI        | Byte | 16(4/0)  | —        | 16(4/0)  |
|             | Word | 16(4/0)  | —        | 16(4/0)  |
|             | Long | 26(6/0)  | —        | 24(6/0)  |
| EORI        | Byte | 16(4/0)  | —        | 20(4/1)+ |
|             | Word | 16(4/0)  | —        | 24(4/2)+ |
|             | Long | 28(6/0)  | —        | 40(6/4)+ |
| MOVEQ       | Long | 8(2/0)   | —        | —        |
| ORI         | Byte | 16(4/0)  | —        | 20(4/1)+ |
|             | Word | 16(4/0)  | —        | 24(4/2)+ |
|             | Long | 28(6/0)  | —        | 40(6/4)+ |
| SUBI        | Byte | 16(4/0)  | —        | 12(2/1)+ |
|             | Word | 16(4/0)  | —        | 16(2/2)+ |
|             | Long | 28(6/0)  | —        | 24(2/4)+ |
| SUBQ        | Byte | 8(2/0)   | —        | 20(4/1)+ |
|             | Word | 8(2/0)   | 12(2/0)  | 24(4/2)+ |
|             | Long | 12(2/0)  | 12(2/0)  | 40(6/4)+ |

+Add effective address calculation time.

## 7.5 SINGLE OPERAND INSTRUCTION EXECUTION TIMES

Table 7-7 lists the timing data for the single operand instructions. The total number of clock periods, the number of read cycles, and the number of write cycles are shown in the previously described format. The number of clock periods, the number of read cycles, and the number of write cycles, respectively, must be added to those of the effective address calculation where indicated by a plus sign (+).

**Table 7-7. Single Operand Instruction Execution Times**

| Instruction | Size        | Register | Memory   |
|-------------|-------------|----------|----------|
| CLR         | Byte        | 8(2/0)   | 12(2/1)+ |
|             | Word        | 8(2/0)   | 16(2/2)+ |
|             | Long        | 10(2/0)  | 24(2/4)+ |
| NBCD        | Byte        | 10(2/0)  | 12(2/1)+ |
| NEG         | Byte        | 8(2/0)   | 12(2/1)+ |
|             | Word        | 8(2/0)   | 16(2/2)+ |
|             | Long        | 10(2/0)  | 24(2/4)+ |
| NEGX        | Byte        | 8(2/0)   | 12(2/1)+ |
|             | Word        | 8(2/0)   | 16(2/2)+ |
|             | Long        | 10(2/0)  | 24(2/4)+ |
| NOT         | Byte        | 8(2/0)   | 12(2/1)+ |
|             | Word        | 8(2/0)   | 16(2/2)+ |
|             | Long        | 10(2/0)  | 24(2/4)+ |
| Scc         | Byte, False | 8(2/0)   | 12(2/1)+ |
|             | Byte, True  | 10(2/0)  | 12(2/1)+ |
| TAS         | Byte        | 8(2/0)   | 14(2/1)+ |
| TST         | Byte        | 8(2/0)   | 8(2/0)+  |
|             | Word        | 8(2/0)   | 8(2/0)+  |
|             | Long        | 8(2/0)   | 8(2/0)+  |

+Add effective address calculation time.

**7**

## 7.6 SHIFT/ROTATE INSTRUCTION EXECUTION TIMES

Table 7-8 lists the timing data for the shift and rotate instructions. The total number of clock periods, the number of read cycles, and the number of write cycles are shown in the previously described format. The number of clock periods, the number of read cycles, and the number of write cycles, respectively, must be added to those of the effective address calculation where indicated by a plus sign (+).

**Table 7-8. Shift/Rotate Instruction Execution Times**

| Instruction | Size | Register   | Memory   |
|-------------|------|------------|----------|
| ASR, ASL    | Byte | 10+2n(2/0) | —        |
|             | Word | 10+2n(2/0) | 16(2/2)+ |
|             | Long | 12+n2(2/0) | —        |
| LSR, LSL    | Byte | 10+2n(2/0) | —        |
|             | Word | 10+2n(2/0) | 16(2/2)+ |
|             | Long | 12+n2(2/0) | —        |
| ROR, ROL    | Byte | 10+2n(2/0) | —        |
|             | Word | 10+2n(2/0) | 16(2/2)+ |
|             | Long | 12+n2(2/0) | —        |
| ROXR, ROXL  | Byte | 10+2n(2/0) | —        |
|             | Word | 10+2n(2/0) | 16(2/2)+ |
|             | Long | 12+n2(2/0) | —        |

+Add effective address calculation time for word operands.  
n is the shift count.

## 7.7 BIT MANIPULATION INSTRUCTION EXECUTION TIMES

Table 7-9 lists the timing data for the bit manipulation instructions. The total number of clock periods, the number of read cycles, and the number of write cycles are shown in the previously described format. The number of clock periods, the number of read cycles, and the number of write cycles, respectively, must be added to those of the effective address calculation where indicated by a plus sign (+).

**Table 7-9. Bit Manipulation Instruction Execution Times**

| Instruction | Size         | Dynamic       |               | Static        |               |
|-------------|--------------|---------------|---------------|---------------|---------------|
|             |              | Register      | Memory        | Register      | Memory        |
| BCHG        | Byte<br>Long | —<br>12(2/0)* | 12(2/1)+<br>— | —<br>20(4/0)* | 20(4/1)+<br>— |
| BCLR        | Byte<br>Long | —<br>14(2/0)* | 12(2/1)+<br>— | —<br>22(4/0)* | 20(4/1)+<br>— |
| BSET        | Byte<br>Long | —<br>12(2/0)* | 12(2/1)+<br>— | —<br>20(4/0)* | 20(4/1)+<br>— |
| BTST        | Byte<br>Long | —<br>10(2/0)  | 8(2/0)+<br>—  | —<br>18(4/0)  | 16(4/0)+<br>— |

+Add effective address calculation time.

\* Indicates maximum value; data addressing mode only.

## 7.8 CONDITIONAL INSTRUCTION EXECUTION TIMES

Table 7-10 lists the timing data for the conditional instructions. The total number of clock periods, the number of read cycles, and the number of write cycles are shown in the previously described format. The number of clock periods, the number of read cycles, and the number of write cycles, respectively, must be added to those of the effective address calculation where indicated by a plus sign (+).

**Table 7-10. Conditional Instruction Execution Times**

| Instruction | Displacement | Trap or Branch<br>Taken | Trap or Branch<br>Not Taken |
|-------------|--------------|-------------------------|-----------------------------|
| Bcc         | Byte         | 18(4/0)                 | 12(2/0)                     |
|             | Word         | 18(4/0)                 | 20(4/0)                     |
| BRA         | Byte         | 18(4/0)                 | —                           |
|             | Word         | 18(4/0)                 | —                           |
| BSR         | Byte         | 34(4/4)                 | —                           |
|             | Word         | 34(4/4)                 | —                           |
| DBcc        | CC True      | —                       | 20(4/0)                     |
|             | CC False     | 18(4/0)                 | 26(6/0)                     |
| CHK         | —            | 68(8/6)+*               | 14(2/0)                     |
| TRAP        | —            | 62(8/6)                 | —                           |
| TRAPV       | —            | 66(10/6)                | 8(2/0)                      |

+Add effective address calculation time for word operand.

\* Indicates maximum base value.

## 7.9 JMP, JSR, LEA, PEA, AND MOVEM INSTRUCTION EXECUTION TIMES

Table 7-11 lists the timing data for the jump (JMP), jump to subroutine (JSR), load effective address (LEA), push effective address (PEA), and move multiple registers (MOVEM) instructions. The total number of clock periods, the number of read cycles, and the number of write cycles are shown in the previously described format.

**Table 7-11. JMP, JSR, LEA, PEA, and MOVEM Instruction Execution Times**

| Instruction    | Size | (An)               | (An)+              | -(An)            | (d <sub>16</sub> ,An) | (d <sub>8</sub> ,An,Xn)+ | (xxx).W            | (xxx).L            | (d <sub>16</sub> PC) | (d <sub>8</sub> , PC, Xn)* |
|----------------|------|--------------------|--------------------|------------------|-----------------------|--------------------------|--------------------|--------------------|----------------------|----------------------------|
| JMP            | —    | 16(4/0)            | —                  | —                | 18(4/0)               | 22(4/0)                  | 18(4/0)            | 24(6/0)            | 18(4/0)              | 22(4/0)                    |
| JSR            | —    | 32(4/4)            | —                  | —                | 34(4/4)               | 38(4/4)                  | 34(4/4)            | 40(6/4)            | 34(4/4)              | 32(4/4)                    |
| LEA            | —    | 8(2/0)             | —                  | —                | 16(4/0)               | 20(4/0)                  | 16(4/0)            | 24(6/0)            | 16(4/0)              | 20(4/0)                    |
| PEA            | —    | 24(2/4)            | —                  | —                | 32(4/4)               | 36(4/4)                  | 32(4/4)            | 40(6/4)            | 32(4/4)              | 36(4/4)                    |
| MOVEM<br>M → R | Word | 24+8n<br>(6+2n/0)  | 24+8n<br>(6+2n/0)  | —                | 32+8n<br>(8+2n/0)     | 34+8n<br>(8+2n/0)        | 32+8n<br>(10+n/0)  | 40+8n<br>(10+2n/0) | 32+8n<br>(8+2n/0)    | 34+8n<br>(8+2n/0)          |
|                | Long | 24+16n<br>(6+4n/0) | 24+16n<br>(6+4n/0) | —                | 32+16n<br>(8+4n/0)    | 34+16n<br>(8+4n/0)       | 32+16n<br>(8+4n/0) | 40+16n<br>(8+4n/0) | 32+16n<br>(8+4n/0)   | 34+16n<br>(8+4n/0)         |
| MOVEM<br>R → M | Word | 16+8n<br>(4/2n)    | —                  | 16+8n<br>(4/2n)  | 24+8n<br>(6/2n)       | 26+8n<br>(6/2n)          | 24+8n<br>(6/2n)    | 32+8n<br>(8/2n)    | —                    | —                          |
|                | Long | 16+16n<br>(4/4n)   | —                  | 16+16n<br>(4/4n) | 24+16n<br>(6/4n)      | 26+16n                   | 24+16n<br>(8/4n)   | 32+16n<br>(6/4n)   | —                    | —                          |

n is the number of registers to move.

\*The size of the index register (Xn) does not affect the instruction's execution time.

## 7.10 MULTIPRECISION INSTRUCTION EXECUTION TIMES

Table 7-12 lists the timing data for multiprecision instructions. The numbers of clock periods include the times to fetch both operands, perform the operations, store the results, and read the next instructions. The total number of clock periods, the number of read cycles, and the number of write cycles are shown in the previously described format.

The following notation applies in Table 7-12:

Dn — Data register operand

M — Memory operand

**Table 7-12. Multiprecision Instruction  
Execution Times**

| Instruction | Size    | op Dn, Dn | op M, M  |
|-------------|---------|-----------|----------|
| ADDX        | Byte    | 8(2/0)    | 22(4/1)  |
|             | Word    | 8(2/0)    | 50(6/2)  |
|             | Long    | 12(2/0)   | 58(10/4) |
| CMPM        | Byte,   | —         | 16(4/0)  |
|             | Word    | —         | 24(6/0)  |
|             | Long    | —         | 40(10/0) |
| SUBX        | Byte, \ | 8(2/0)    | 22(4/1)  |
|             | Word    | 8(2/0)    | 50(6/2)  |
|             | Long    | 12(2/0)   | 58(10/4) |
| ABCD        | Byte    | 10(2/0)   | 20(4/1)  |
| SBCD        | Byte    | 10(2/0)   | 20(4/1)  |

## 7.11 MISCELLANEOUS INSTRUCTION EXECUTION TIMES

Tables 7-13 and 7-14 list the timing data for miscellaneous instructions. The total number of clock periods, the number of read cycles, and the number of write cycles are shown in the previously described format. The number of clock periods, the number of read cycles, and the number of write cycles, respectively, must be added to those of the effective address calculation where indicated by a plus sign (+).

**Table 7-13. Miscellaneous Instruction Execution Times**

| Instruction     | Register | Memory   |
|-----------------|----------|----------|
| ANDI to CCR     | 32(6/0)  | —        |
| ANDI to SR      | 32(6/0)  | —        |
| EORI to CCR     | 32(6/0)  | —        |
| EORI to SR      | 32(6/0)  | —        |
| EXG             | 10(2/0)  | —        |
| EXT             | 8(2/0)   | —        |
| LINK            | 32(4/4)  | —        |
| MOVE to CCR     | 18(4/0)  | 18(4/0)+ |
| MOVE to SR      | 18(4/0)  | 18(4/0)+ |
| MOVE from SR    | 10(2/0)  | 16(2/2)+ |
| MOVE to USP     | 8(2/0)   | —        |
| MOVE from USP   | 8(2/0)   | —        |
| NOP             | 8(2/0)   | —        |
| ORI to CCR      | 32(6/0)  | —        |
| ORI to SR       | 32(6/0)  | —        |
| RESET           | 136(2/0) | —        |
| RTE             | 40(10/0) | —        |
| RTR             | 40(10/0) | —        |
| RTS             | 32(8/0)  | —        |
| STOP            | 4(0/0)   | —        |
| SWAP            | 8(2/0)   | —        |
| TRAPV (No Trap) | 8(2/0)   | —        |
| UNLK            | 24(6/0)  | —        |

+Add effective address calculation time for word operand.

**Table 7-14. Move Peripheral Instruction Execution Times**

| Instruction | Size | Register → Memory | Memory → Register |
|-------------|------|-------------------|-------------------|
| MOVEP       | Word | 24(4/2)           | 24(6/0)           |
|             | Long | 32(4/4)           | 32(8/0)           |

+Add effective address calculation time.

## 7.12 EXCEPTION PROCESSING EXECUTION TIMES

Table 7-15 lists the timing data for exception processing. The numbers of clock periods include the times for all stacking, the vector fetch, and the fetch of the first instruction of the handler routine. The total number of clock periods, the number of read cycles, and the number of write cycles are shown in the previously described format. The number of clock

periods, the number of read cycles, and the number of write cycles, respectively, must be added to those of the effective address calculation where indicated by a plus sign (+).

**Table 7-15. Exception Processing  
Execution Times**

| <b>Exception</b>    | <b>Periods</b> |
|---------------------|----------------|
| Address Error       | 94(8/14)       |
| Bus Error           | 94(8/14)       |
| CHK Instruction     | 68(8/6)+       |
| Divide by Zero      | 66(8/6)+       |
| Interrupt           | 72(9/6)*       |
| Illegal Instruction | 62(8/6)        |
| Privilege Violation | 62(8/6)        |
| RESET**             | 64(12/0)       |
| Trace               | 62(8/6)        |
| TRAP Instruction    | 62(8/6)        |
| TRAPV Instruction   | 66(10/6)       |

+ Add effective address calculation time.

\*\* Indicates the time from when RESET and HALT are first sampled as negated to when instruction execution starts.

